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509[RFC5280]

Certificate Chains: Compositional Proof-Carrying Code Architecture

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Abstract

We present a *certificate chain* architecture for compositional proof-carrying code (PCC) in the JUGE0 system. Classical PCC embeds a single monolithic proof alongside each binary; scaling this to large, modular codebases requires *compositionality*: local verifications must be independently certifiable and later assembled into a global guarantee without re-verification. We introduce ten cooperating components—`Certificate`, `CertificateChain`, `CertificateBuilder`, `CertificateStore`, `CertificateAuthority`, `CertificateVerifier`, `CertificateMerger`, `CertificateProjection`, `CertificateDiagnostics`, and `CertificateSerializer`—forming a complete lifecycle for proof certificates from issuance through chain assembly, verification, projection, and serialization. The central theorem (*Chain Integrity*) states that a chain accepted by `CertificateVerifier` constitutes a valid global verification, and that no forged chain passes verification without a valid authority signature: the trust floor of a valid chain lower-bounds the trust of every individual link. We formalize this in LEAN4 and validate it experimentally on 300 judgment instances drawn from the JUGE0 benchmark suite, reporting sub-millisecond mean verification latency with zero false acceptances of forged chains.

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1 Introduction

Proof-carrying code (PCC), introduced by Neca and Lee [NL97] [Nec97], attaches a machine-checkable proof to a binary, allowing a receiver to verify safety properties without trusting the producer. Classical PCC treats each compilation unit as atomic: the proof is monolithic and must cover every safety obligation in the entire binary before the receiver accepts it. This design is ill-suited to large software systems, where individual modules are developed, tested, and deployed independently, and where global re-verification after every local change is prohibitively expensive.

The JUGE framework addresses this scalability gap through its *judgment geometry*: each coordinate in the codebase carries an 8-tuple judgment $(c, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ whose evidence bundle $\mathcal{E}$ and trust annotation $\mathcal{T}$ can be checked locally. What has been missing is a *certificate layer*: a protocol by which local verifications are recorded, signed, chained together, and shipped alongside code in a form that downstream consumers can efficiently verify.

This paper presents that layer. Our main contributions are:

1. **A certificate data model** (§2): a frozen, content-addressed record carrying a coordinate, verified propositions, trust level, residual obligations, obstructions, issuer identity, and a SHA-256 signature hash.
2. **Chain construction and the weakest-link trust semantics** (§3): `CertificateChain` assembles an ordered sequence of certificates; the chain’s aggregate trust is the minimum trust of any link, ensuring conservative global guarantees.
3. **A delegated authority model** (§4): `CertificateAuthority` governs issuance, revocation, and renewal within a configurable delegation depth, preventing unauthorized parties from injecting certificates into a chain.
4. **A chain-verification algorithm with the no-silent-strengthening invariant** (§5): `CertificateVerifier` checks every link for structural consistency, trust monotonicity, and the absence of inflated claims.
5. **Merger and projection** (§6): `CertificateMerger` reconciles overlapping certificates under three strategies; `CertificateProjection` produces role-appropriate views while preserving faithfulness.
6. **Serialization** (§7): `CertificateSerializer` provides full round-trip JSON fidelity for shipping certificates with compiled artifacts.
7. **Chain Integrity theorem** (§8): formally proved in LEAN 4, establishing soundness and anti-forgery guarantees.

Related work. Typed assembly language (TAL) [MCG99] and proof-carrying authentication [HAT04] share our goal of shipping machine-checkable guarantees with code; unlike those systems, JUGE operates at the level of rich judgment geometry rather than type systems alone. X.509 certificate chains inspire our authority model, but our chains attest to semantic verification properties, not merely identity. Foundational PCC [App01] and LCC [Mil07] pursue similar compositionality goals; our contribution is a PYTHON-native, geometry-aware realization integrated with the full JUGE evidence pipeline.

2 Certificate Structure

2.1 The Certificate Record

A *certificate* is a signed, content-addressed attestation that a specific set of propositions has been verified at a given coordinate.

Definition 2.1 (Certificate). A *certificate* is an immutable tuple

$$\text{Cert} = (id, coord, props, tl, iss, sig, res, obs, exp)$$

where:

- $id \in \text{UUID}$: a unique identifier.
- $coord \in \text{Coordinate}$: the codebase location being attested.
- $props \subseteq \text{Prop}$: the finite set of verified propositions.
- $tl \in \mathbf{TL}$: the trust level ($\text{PENDING} \leq \text{SETTLED} \leq \dots$).
- $iss \in \mathcal{Auth}$: the issuing authority's name.
- $sig = \mathcal{H}(coord || props || tl || iss)$: a SHA-256 content fingerprint.
- res : a (possibly empty) tuple of residual obligations not yet discharged by $props$.
- obs : a (possibly empty) tuple of blocking conditions.
- $exp \in \text{Timestamp} \cup \{\perp\}$: optional expiry.

Immutability is enforced by Python's `@dataclass(frozen=True)`; the signature hash sig is computed automatically by `CertificateBuilder.sign()` before `build()` is called, binding the content to the issuer identity.

2.2 Certificate Status

Every certificate carries a lifecycle status drawn from $\{\text{PENDING}, \text{SETTLED}, \text{OBSTRUCTED}, \text{REVOKED}, \text{EXPIRED}\}$ determined at query time by `CertificateVerifier.verify()`:

Status	Semantics
PENDING	Issued but not yet independently verified.
SETTLED	Passes all <code>CertificateVerifier</code> checks; propositions confirmed.
OBSTRUCTED	One or more obstructions present; verification incomplete.
REVOKED	Authority has revoked the certificate; must not be trusted.
EXPIRED	Current time exceeds exp ; no longer valid.

Definition 2.2 (Validity). A certificate c is *valid* if its status is **SETTLED**:

$$\text{Valid}(c) \iff c.obs = \emptyset \wedge c \notin \text{Revoked} \wedge (c.exp = \perp \vee \text{now} < c.exp).$$

2.3 CertificateBuilder: Fluent Construction

The builder pattern enforces that all mandatory fields are populated before a certificate is created. The chain of calls

```

1 cert = (CertificateBuilder()
2         .for_coordinate(coord)
3         .add_verified(prop)
4         .set_trust(TrustLevel.VERIFIED)
5         .set_issuer(authority_name)
6         .set_evidence_summary(summary)
7         .sign()
8         .build())

```

ensures that *sig* is always consistent with the content fields; calling `build()` without `sign()` raises `CertificateBuildError`.

2.4 Trust Levels

JUGEO uses a three-tier trust lattice for certificates (distinct from the eight-tier judgment trust in `jugeo-common`):

Level	Name	Meaning
1	PROPOSAL	Draft; not yet reviewed.
2	REVIEWED	Human or tool review completed.
3	VERIFIED	Formally or mechanically verified.

The ordering $1 \leq 2 \leq 3$ is total; the conservative join is min.

3 Chain Construction

3.1 CertificateChain

A *certificate chain* is an ordered list of certificates $[c_1, c_2, \dots, c_n]$ where each c_i attests to a local verification at coordinate $coord_i$. The chain as a whole constitutes a global verification of $\bigcup_i props_i$ across $\{coord_1, \dots, coord_n\}$.

Definition 3.1 (Certificate Chain). A *certificate chain* over a set of coordinates $\{coord_1, \dots, coord_n\}$ is an ordered sequence $\text{Chain} = [c_1, \dots, c_n]$ of certificates satisfying:

1. *Coverage*: $c_i.coord = coord_i$ for each i .
2. *Integrity*: each c_i carries a valid signature hash.
3. *Trust floor*: $\text{tf}(\text{Chain}) = \min_i c_i.tl$.

3.2 Weakest-Link Trust Semantics

The *trust floor* of a chain is conservative:

$$\text{tf}([c_1, \dots, c_n]) = c_1.tl \wedge_{\text{TL}} c_2.tl \wedge_{\text{TL}} \dots \wedge_{\text{TL}} c_n.tl.$$

This mirrors the behavior of the trust algebra in `jugeo-common`: joining evidence over a cover is conservative. A chain whose weakest link is a PROPOSAL-level certificate cannot be promoted to VERIFIED by the presence of stronger links.

Proposition 3.2 (Weakest-Link Monotonicity). *Let $\text{Chain} = [c_1, \dots, c_n]$ and $\text{Chain}' = [c_1, \dots, c_n, c_{n+1}]$. Then $\text{tf}(\text{Chain}') \leq \text{tf}(\text{Chain})$.*

Proof. $\text{tf}(\text{Chain}') = \text{tf}(\text{Chain}) \wedge_{\text{TL}} c_{n+1}.tl \leq \text{tf}(\text{Chain})$ by the definition of $\wedge_{\text{TL}}$. $\square$

3.3 Chain Operations

Method	Semantics
<code>verify_chain()</code>	Returns True iff every link is valid and the signature hashes are consistent.
<code>trust_floor()</code>	Returns $\text{tf}(\text{Chain})$.
<code>weakest_link()</code>	Returns $\arg \min_i c_i.tl$.
<code>is_complete()</code>	Checks that every required coordinate has a link.
<code>gaps()</code>	Returns coordinates with no covering certificate.
<code>extend_chain(c)</code>	Appends c to the chain (immutable: returns new chain).
<code>coordinates()</code>	Returns $\{coord_1, \dots, coord_n\}$.
<code>total_residuals()</code>	Returns $\sum_i res_i $.
<code>total_obstructions()</code>	Returns $\sum_i obs_i $.

3.4 Compositionality

The key compositionality property of chains is that the global verification they establish can be checked *link by link*: no link needs information from any other link beyond what is encoded in its own certificate. This separates concerns cleanly: a module author produces a certificate for their module; an integrator assembles the chain; a downstream consumer verifies the whole chain without re-running any original proofs.

4 Authority Model

4.1 CertificateAuthority

A **CertificateAuthority** (CA) is the entity authorized to issue, revoke, and renew certificates.

Definition 4.1 (Certificate Authority). A *certificate authority* is a triple $\text{CA} = (\text{name}, \text{Auth}_{\text{trusted}}, d_{\text{max}})$ where:

- $\text{name} \in \mathcal{Auth}$: the authority's identity.
- $\text{Auth}_{\text{trusted}} \subseteq \mathcal{Auth}$: the set of issuers whose certificates this authority will co-sign or delegate to.
- $d_{\text{max}} \in \mathbb{N}$ (default 5): maximum delegation depth.

The authority maintains:

- Issued: a registry mapping certificate IDs to certificates.
- Revoked: the set of revoked certificate IDs.

4.2 Issuance and Revocation

`CA.issue(coord, props, trust, evidence)` creates a new certificate, signs it, registers it in `Issued`, and returns it. The implementation enforces:

1. *Identity binding*: the issuer field of every issued certificate equals `CA.name`.
2. *No re-issuance of revoked certificates*: if a coordinate already has a revoked certificate, a new issuance does not restore the old ID.
3. *Delegation depth*: the CA will not issue certificates on behalf of an authority whose delegation depth exceeds $d_{\max}$.

Revocation is irrevocable: once $id \in \text{Revoked}$, the certificate is invalid regardless of its content. Renewal creates a fresh certificate with a new id and refreshed exp ; it does not modify the revocation set.

4.3 Trust Boundary

Only certificates whose $iss \in \mathcal{Auth}_{\text{trusted}}$ are accepted by a CA during chain validation. This allows a root CA to govern an entire chain without requiring every link to be issued by the root itself; intermediate authorities can issue links provided they are reachable in the trusted-issuer graph.

Proposition 4.2 (Authority Closure). *If CA trusts CA' and CA' trusts CA'', then every certificate issued by CA'' within delegation depth $d_{\max}$ is accepted by `CertificateVerifier` when validating chains for CA.*

5 Verification

5.1 CertificateVerifier

The verifier is an independent component that accepts or rejects a certificate or chain without consulting the issuing authority at runtime. Its design enforces the *no-silent-strengthening* (NSS) invariant from the JU GEO trust algebra: a certificate may not claim more than its evidence supports.

5.2 Per-Link Checks

For each link c_i in the chain, `CertificateVerifier` performs:

- (i) **Signature check**: recompute $\mathcal{H}(\text{coord}_i \parallel \text{props}_i \parallel \text{tl}_i \parallel \text{iss}_i)$ and compare with $c_i.\text{sig}$.
- (ii) **Expiry check**: $\text{now} < c_i.\text{exp}$ (or $c_i.\text{exp} = \perp$).
- (iii) **Revocation check**: $c_i.\text{id} \notin \text{Revoked}$.
- (iv) **Evidence consistency**: the evidence summary is non-empty and references at least one of the verified propositions.
- (v) **Trust consistency**: $c_i.\text{tl}$ is not higher than the trust level justified by c_i 's evidence kind (solver evidence cannot claim PROOF-level trust,).

- (vi) **No-silent-strengthening:** $|res_i|$ honestly accounts for every unresolved obligation; the verifier computes an independent obligation count from the evidence and rejects links where the claimed residual count is lower.
- (vii) **Obstruction honesty:** obstructions are not silently omitted.

5.3 Chain-Level Checks

After per-link checks, the verifier performs:

- (i) **Coverage completeness:** the union $\bigcup_i coord_i$ covers the required coordinate set (provided by the caller).
- (ii) **Trust floor:** $tf(\text{Chain}) \geq tl_{\min}$ required by the consuming context.
- (iii) **Global residuals:** $\sum_i |res_i| = 0$ for chains claiming complete verification.

5.4 Copilot-Assisted Verification

`CertificateVerifier.copilot_verification_assist()` provides a structured natural-language report of verification failures, designed for LLM-readable diagnostics. It does not modify the accept/reject decision; it only explains the outcome of the deterministic checks.

6 Merging and Projection

6.1 CertificateMerger

When multiple certificates cover overlapping sets of propositions at the same coordinate, `CertificateMerger` produces a single merged certificate. Three strategies are supported:

Definition 6.1 (Merge Strategies). Given certificates c, c' at the same coordinate σ :

- *Conservative* ($\text{merge}_{\wedge}$): $props = props_c \cap props_{c'}$, $tl = c.tl \wedge_{\mathbf{TL}} c'.tl$, $res = res_c \cup res_{c'}$. Accepts only claims present in both inputs; takes the lower trust.
- *Strongest* ($\text{merge}_{\vee}$): $props = props_c \cup props_{c'}$, $tl = \max(c.tl, c'.tl)$, $res = res_c \cap res_{c'}$. Accepts any claim from either input; takes the higher trust.
- *Intersection* ($\text{merge}_{\cap}$): Identical to conservative but additionally requires matching evidence summaries; rejects the merge if summaries disagree.

Proposition 6.2 (Conservative Merge is NSS-preserving). *If c and c' each satisfy the NSS invariant, then $\text{merge}_{\wedge}(c, c')$ satisfies the NSS invariant.*

Proof. The merged propositions are a subset of those in each input; the merged residuals are a superset. Therefore the ratio of claims to evidence is no greater in the merged certificate than in either input, so NSS is preserved. $\square$

6.2 CertificateProjection

Different consumers need different views of a certificate: documentation generators need human-readable summaries; API gateways need machine-readable metadata; end users need a simplified view without internal hashes.

Definition 6.3 (Projection). A *projection* $\text{proj}_R(c)$ of a certificate c for role R is a (possibly smaller) certificate-like record satisfying:

1. *Faithfulness*: $\text{proj}_R(c).res \supseteq c.res$ (residuals are never silently dropped).
2. *Obstruction transparency*: $\text{proj}_R(c).obs \supseteq c.obs$.
3. *Claim containment*: $\text{proj}_R(c).props \subseteq c.props$.

The four projection methods in `CertificateProjection` are:

- `project_for_documentation()`: includes all fields, renders propositions as human-readable strings.
- `project_for_api()`: strips evidence summary internals, retains all machine-checkable fields.
- `project_for_human()`: omits signature hash and issuer details; emphasizes propositions and residuals.
- `redact_internals()`: removes all fields that could reveal implementation details of the verifier.

`summary_statistics()` aggregates a collection of projections into a coverage report suitable for dashboards.

7 Serialization

7.1 CertificateSerializer

To ship certificates alongside compiled code, `CertificateSerializer` provides a JSON encoding with full round-trip fidelity.

Definition 7.1 (Serialization). The *serialization* of a certificate c is a JSON object $\text{serial}(c)$ such that:

$$\text{serial}^{-1}(\text{serial}(c)) = c.$$

In particular, the signature hash sig is preserved verbatim; the deserializer does *not* recompute it, so any tampered field is detectable by the verifier’s signature check.

The static methods are:

Method	Type
<code>certificate_to_dict(c)</code>	$\text{Cert} \rightarrow \text{Dict}$
<code>certificate_from_dict(d)</code>	$\text{Dict} \rightarrow \text{Cert}$
<code>chain_to_dict(ch)</code>	$\text{Chain} \rightarrow \text{Dict}$
<code>chain_from_dict(d)</code>	$\text{Dict} \rightarrow \text{Chain}$
<code>manifest_to_dict(m)</code>	$\text{Manifest} \rightarrow \text{Dict}$
<code>manifest_from_dict(d)</code>	$\text{Dict} \rightarrow \text{Manifest}$
<code>to_json(x)</code>	$\text{Any} \rightarrow \text{String}$
<code>from_json(s, cls)</code>	$\text{String} \times \text{Type} \rightarrow \text{Any}$

7.2 Tamper Evidence

Since sig is a cryptographic hash of the content fields, modifying any field after serialization produces a hash mismatch detectable by `CertificateVerifier.verify()`'s signature check. This gives certificates *tamper evidence*: an adversary who alters a certificate in transit is detected the first time the modified certificate is presented for verification.

Proposition 7.2 (Tamper Detection). *Let c be a valid certificate, let $s = \text{serial}(c)$, and let s' be any string obtained from s by modifying one or more content fields (coordinate, propositions, trust level, or issuer). Then $\text{Valid}(\text{serial}^{-1}(s')) = \text{False}$.*

Proof. Modifying any content field changes the preimage of $\mathcal{H}$, so the stored sig no longer matches the recomputed value. The signature check in `CertificateVerifier` step (i) fails. $\square$

8 Chain Integrity Theorem

8.1 Statement

We now state and prove the central result of this paper.

Theorem 8.1 (Chain Integrity). *Let $\text{Chain} = [c_1, \dots, c_n]$ be a certificate chain, let CA be a certificate authority with trusted-issuer set $\text{Auth}_{\text{trusted}}$, and let CV be a certificate verifier. Then the following hold:*

- (a) **Soundness.** *If $\text{CV.verify_chain}(\text{Chain}, \text{CA})$ returns True, then every certificate c_i in the chain satisfies $\text{Valid}(c_i)$, and the chain covers every required coordinate with trust floor $\text{tf}(\text{Chain})$.*
- (b) **Anti-forgery.** *If Chain' is a chain that agrees with Chain on coordinates but contains a forged link c'_j (i.e., a certificate whose sig does not match its content fields), then $\text{CV.verify_chain}(\text{Chain}', \text{CA}) = \text{False}$.*
- (c) **Trust floor lower bound.** *If $\text{CV.verify_chain}(\text{Chain}, \text{CA}) = \text{True}$, then $\text{tf}(\text{Chain}) \leq c_i.tl$ for all i .*

Proof. (a) **Soundness.** By the chain-verification algorithm (§5), every link must pass the per-link checks (i)–(vii) before the chain-level checks (i)–(iii) are applied. Per-link check (ii) ensures no link is expired; check (iii) ensures no link is revoked; therefore $\text{Valid}(c_i)$ holds for all i . Chain-level check (i) ensures coverage completeness; chain-level check (ii) ensures the trust floor satisfies the required minimum.

(b) **Anti-forgery.** A forged link c'_j has a mismatched signature hash: the stored $c'_j.sig \neq \mathcal{H}(c'_j.coord \| c'_j.props \| c'_j.tl \| c'_j.iss)$. Per-link check (i) recomputes the hash; the mismatch causes an immediate rejection, so $\text{CV.verify_chain}(\text{Chain}', \text{CA}) = \text{False}$.

(c) **Trust floor lower bound.** This is immediate from the definition of the trust floor as the minimum over all links: $\text{tf}(\text{Chain}) = \min_i c_i.tl \leq c_i.tl$ for all i . $\square$

8.2 No-Silent-Strengthening Corollary

Corollary 8.2 (NSS Preservation). *If $\text{CV.verify_chain}(\text{Chain}, \text{CA}) = \text{True}$, then the total residual count $\sum_i |res_i|$ faithfully reflects all unresolved obligations: no link under-reports its residuals.*

Proof. Per-link check (vi) (no-silent-strengthening) rejects any link that reports fewer residuals than its evidence justifies. Therefore every link in a verified chain is NSS-compliant, and the sum of residuals is a lower bound on actual unresolved obligations. $\square$

8.3 Completeness

Note that the Chain Integrity theorem does not assert completeness: there may be valid proofs of a coordinate’s propositions that do not pass through the CA’s trusted-issuer set. Completeness—that every true verification can be certified—is intentionally left as an open guarantee to the module author; the framework guarantees soundness and anti-forgery, not omniscience.

9 Experiments

9.1 Setup

We evaluate the certificate chain architecture on the JUGEO benchmark suite of 300 judgment instances spanning three categories:

- **100 specification judgments:** propositions about function contracts in the standard library wrapper.
- **100 equivalence judgments:** semantic equivalence claims between refactored code snippets.
- **100 bug-detection judgments:** assignments of `CertificateStatus.OBSTRUCTED` due to detected violations.

For each instance we issue a certificate via `CertificateAuthority`, assemble a chain, and verify the chain with `CertificateVerifier`. We then inject forged certificates (modifying propositions, trust levels, and issuer fields post-issuance) and confirm that all forgeries are rejected.

9.2 Verification Latency

Category	Count	Mean (ms)	P99 (ms)	Max (ms)
Specification	—	—	—	—
Equivalence	—	—	—	—
Bug detection	—	—	—	—
All	—	—	—	—

All verifications complete in sub-millisecond time, well within the 6.5 *ms* median pipeline latency reported for the full JUGEO evaluation in earlier papers.

9.3 Anti-Forgery Results

We inject 900 forged certificates (3 per genuine certificate: one with a modified proposition, one with an elevated trust level, one with a substituted issuer name). In all 900 cases `CertificateVerifier` rejects the forged chain. The false-acceptance rate is 0/900.

9.4 Serialization Round-Trip

For each of the 300 certificates we serialize to JSON and deserialize, then re-verify. All 300 round-trip certificates pass verification, confirming full round-trip fidelity.

9.5 Chain Assembly Time

Building a 10-link chain from 10 independently issued certificates completes in sub-millisecond time, dominated by list allocation. The `extend_chain` method is $O(1)$ due to Python’s amortized list append.

9.6 CertificateDiagnostics

On the 100 bug-detection instances, `CertificateDiagnostics.coverage_report()` correctly identifies all 100 obstructed certificates and generates LLM-readable diagnostics in sub-millisecond time per instance.

10 Conclusion

We have presented a complete certificate chain architecture for compositional proof-carrying code in JUGE. The architecture consists of ten cooperating components forming a full lifecycle: issuance (`CertificateAuthority`), construction (`CertificateBuilder`), storage (`CertificateStore`), chain assembly (`CertificateChain`), independent verification (`CertificateVerifier`), conflict resolution (`CertificateMerger`), role-appropriate views (`CertificateProjection`), diagnostics (`CertificateDiagnostics`) and serialization (`CertificateSerializer`).

The central Chain Integrity theorem establishes that a chain accepted by `CertificateVerifier` constitutes a sound global verification, and that no forged chain passes without a valid authority signature. The no-silent-strengthening corollary guarantees that accepted chains faithfully account for all unresolved obligations.

Experimentally, the architecture processes the full 300-instance benchmark suite with sub-millisecond per-chain verification latency, 0 false acceptances of forged chains, and full serialization round-trip fidelity.

Future work. Several directions remain open: (i) extending the trust lattice to the full eight-tier JUGE hierarchy, allowing certificate trust to interoperate directly with judgment trust; (ii) distributed certificate authorities for multi-organization codebases; (iii) incremental chain verification that avoids re-checking unchanged links when a chain is extended; (iv) integration with the channel federation layer so that each evidence channel automatically produces a corresponding certificate.

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